

PROJECT FILE

Customer Churn Predictor System Using Spring Boot and Power BI



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1. ABSTRACT –

Customer churn refers to the phenomenon where customers stop using a company's products or services. It is a major concern for organizations, especially in sectors like telecommunications, banking, and subscription-based platforms, where customer retention directly impacts revenue and growth. Predicting churn in advance enables businesses to take proactive actions such as targeted marketing, personalized offers, and improved customer service. This project focuses on developing a Customer Churn Prediction System that uses machine learning techniques to identify customers who are likely to leave.

The system integrates multiple technologies to achieve both prediction and visualization. Machine learning is implemented using Python, where a classification model is trained on historical customer data. Important features such as tenure, monthly charges, total charges, gender, and contract type are used to build the model. The trained model outputs a churn probability score, which helps classify customers into different risk categories such as high risk or low risk.

For backend processing, Flask is used to deploy the machine learning model as an API, while Spring Boot is implemented as an integration layer to demonstrate communication between different services. The system processes user inputs, generates predictions, and stores the results in a dataset. Power BI is used as the visualization tool to create interactive dashboards that display key insights, trends, and prediction results in an easy-to-understand format.

The Power BI dashboard provides both analytical and predictive views. It includes visualizations such as churn rate analysis, customer segmentation, and risk distribution. Additionally, a prediction page allows users to input customer details and view predicted churn probability along with meaningful insights.

Key Features of the System:

- Machine learning-based churn prediction using classification algorithms
- Backend integration using Flask and Spring Boot APIs
- Interactive Power BI dashboard for data visualization
- Real-time simulation of predictions through dataset updates
- User-friendly interface for analyzing customer behaviour

The proposed system demonstrates the importance of data-driven decision-making in improving customer retention. It provides a scalable and flexible framework that can be extended to real-time systems, web applications, or enterprise-level deployments. By combining machine learning with business intelligence tools, the project highlights an effective approach to solving real-world problems related to customer churn.

2. INTRODUCTION –

- **Evolution of Data-Driven Decision Making –**

In recent years, businesses have shifted from traditional decision-making methods to data-driven approaches. With the increasing availability of customer data, organizations are now able to analyze patterns and trends to improve their strategies. Predictive analytics plays a crucial role in this transformation by enabling companies to anticipate future outcomes based on historical data.

- **Challenges in Customer Behaviour Analysis –**

Understanding customer behaviour is complex due to the large volume and variety of data involved. Customers interact with services in different ways, and their decisions are influenced by multiple factors such as pricing, service quality, and engagement levels. Identifying meaningful patterns from such data requires advanced analytical techniques.

- **Role of Predictive Analytics in Business Strategy –**

Predictive analytics helps organizations move from reactive to proactive strategies. Instead of responding after customers leave, businesses can identify early warning signs and take preventive actions. This improves customer satisfaction and reduces financial losses.

- **Application of Machine Learning –**

Machine learning algorithms are widely used for classification and prediction tasks. These algorithms learn from past data and identify relationships between different features. In this project, machine learning is used to classify customers based on their likelihood of churn, providing a probability score that aids in decision-making.

- **Integration of Multiple Technologies –**

Modern systems often combine multiple technologies to achieve efficient results. In this project, Python is used for model development, Flask is used for deploying the model, Spring Boot demonstrates backend integration, and Power BI is used for visualization. This integration ensures a complete end-to-end system.

- **Importance of Visualization Tools –**

Raw data and predictions are difficult to interpret without proper visualization. Tools like Power BI help convert complex data into interactive dashboards with charts and indicators. This allows users to quickly understand patterns, trends, and predictions without needing technical expertise.

- **System-Oriented Approach –**

The project follows a system-based approach where different components work together to achieve a common goal. Data is processed, predictions are generated, and results are presented visually. This structured approach improves efficiency and usability.

- **Practical Relevance of the Project –**

The concepts used in this project are widely applicable in real-world scenarios. Organizations can use similar systems to monitor customer behavior, reduce churn, and improve business performance. The project demonstrates how theoretical knowledge can be applied to solve practical problems.

- **Scalability and Adaptability –**

The system is designed in a modular way, allowing future enhancements such as real-time prediction, integration with web applications, and deployment in cloud environments. This makes the project adaptable to different business requirements.

3. PROBLEM STATEMENT –

- **Customer Retention Challenge –**
Organizations face significant challenges in retaining customers due to increasing competition and changing customer expectations. Losing customers not only affects revenue but also increases the cost of acquiring new customers.
- **Lack of Predictive Capability in Traditional Systems –**
Many existing systems rely on historical reporting and do not provide predictive insights. They identify churn only after it has occurred, which limits the ability of businesses to take preventive action.
- **Complexity of Customer Data –**
Customer data is large, diverse, and continuously growing. Extracting meaningful insights from such data manually or using basic tools is inefficient and prone to errors.
- **Need for Integrated Solution –**
There is a need for a system that can combine data processing, prediction, and visualization in a unified manner. Such a system should be capable of handling inputs, generating predictions, and presenting results in an understandable format.
- **Problem Definition –**
The problem addressed in this project is to design and develop a system that can analyze customer data, predict the likelihood of churn, and provide insights through an interactive visualization platform.

4. OBJECTIVES –

- **To Analyze Customer Data –**
Study and understand customer-related features such as tenure, charges, and contract type to identify factors influencing churn.
- **To Develop a Machine Learning Model –**
Build a classification model capable of predicting whether a customer is likely to churn based on input features.
- **To Generate Churn Probability –**
Provide a probability score that indicates the risk level of each customer, enabling better decision-making.
- **To Implement Backend Integration –**
Develop backend services using Flask and demonstrate integration with Spring Boot to handle prediction requests.
- **To Visualize Data Using Power BI –**
Create interactive dashboards that display churn trends, customer segmentation, and prediction results.
- **To Design a User-Friendly Interface –**
Ensure that the system is easy to use and allows users to view predictions and insights without technical complexity.
- **To Simulate Real-World Application –**

Demonstrate how machine learning and data visualization can be applied in real business scenarios for customer retention.

- **To Enable Data-Driven Decision Making –**

Provide insights that help organizations take proactive measures to reduce churn and improve customer satisfaction.

5. SYSTEM ARCHITECTURE –

- **Overview of the Architecture –**

The Customer Churn Prediction System follows a modular architecture where different components work together to process data, generate predictions, and visualize results. The system is designed to ensure smooth communication between the machine learning model, backend services, and visualization layer.

- **Architecture Components –**

The system consists of the following major components:

- **Input Layer (User Input / API Request) –**
This layer is responsible for receiving input data such as tenure, monthly charges, total charges, gender, and contract type. The input can be provided through an API request (Postman or application interface).
- **Backend Layer (Spring Boot + Flask) –**
The backend layer handles communication between the input and the machine learning model.
 - Spring Boot acts as an integration layer that receives requests and forwards them to the Flask API.
 - Flask serves as the model deployment layer, where the trained machine learning model is hosted and executed.
- **Machine Learning Model Layer –**
This layer contains the trained classification model developed using Python. The model processes input features and generates a churn probability score, which is then used to classify the customer as high risk or low risk.
- **Data Storage Layer (CSV Files) –**
The prediction results are stored in CSV files such as `live_prediction.csv`. This acts as a bridge between the backend system and the visualization tool.
- **Visualization Layer (Power BI) –**
Power BI is used to create interactive dashboards. It reads data from the CSV files and displays insights such as churn probability, risk levels, customer segmentation, and trends.

- **System Workflow –**

The system operates through the following sequence of steps:

1. User provides input data (customer details)
2. Spring Boot receives the request and forwards it to Flask API
3. Flask processes the input using the trained machine learning model
4. The model generates churn probability and prediction
5. Results are stored in a CSV file
6. Power BI reads the updated data

7. Dashboard is refreshed to display new predictions

- **Architecture Flow Representation –**

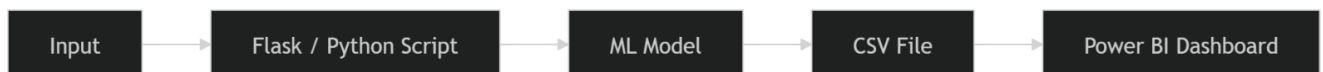


- **Design Advantages –**

- a) Modular structure allows easy maintenance and updates
- b) Separation of concerns between backend, model, and visualization
- c) Scalable design that can support real-time systems
- d) Easy integration with external applications and APIs

- **Alternative Execution Mode (Simplified) –**

For demonstration and offline execution, the system can operate without Spring Boot:



This ensures reliability during testing and practical demonstration while maintaining the same core functionality.

6. METHODOLOGY –

The methodology describes the step-by-step process followed to design and implement the Customer Churn Prediction System. It includes data handling, model development, prediction generation, and visualization.

- **Data Collection –**

The first step involves collecting the dataset required for training the model. The dataset contains customer information such as tenure, monthly charges, total charges, gender, and contract type. This data is essential for understanding customer behaviour and identifying patterns related to churn.

- **Data Preprocessing –**

Raw data is often incomplete or inconsistent, so preprocessing is required to make it suitable for machine learning. The following steps are performed:

- Handling missing or null values
- Converting categorical variables into numerical format (encoding)
- Scaling or normalizing numerical features if required
- Removing irrelevant or duplicate data

This step ensures that the dataset is clean and ready for model training.

- **Feature Selection –**

Important features that influence customer churn are selected. These include:

- Tenure (duration of customer relationship)
- Monthly Charges
- Total Charges
- Gender

- Contract Type

Selecting relevant features improves model accuracy and efficiency.

- **Model Training –**

A machine learning classification model is trained using the processed dataset. Algorithms such as Logistic Regression or Random Forest can be used for this purpose. The model learns patterns from historical data and identifies relationships between input features and churn outcomes.

- **Model Evaluation –**

After training, the model is evaluated to check its performance. Metrics such as accuracy, precision, and recall are used to measure how well the model predicts churn. This step ensures the reliability of the model before deployment.

- **Model Deployment (Flask API) –**

The trained model is deployed using Flask, which acts as an API. This allows external systems to send input data and receive predictions. The API processes incoming requests, applies the model, and returns churn probability and prediction results.

- **Backend Integration (Spring Boot) –**

Spring Boot is used to demonstrate backend integration. It acts as an intermediate layer that receives user requests and forwards them to the Flask API. This simulates how enterprise applications interact with machine learning services.

- **Prediction Generation –**

When input data is provided, the system processes it through the model and generates:

- Churn probability (a numerical value)
- Prediction label (High Risk / Low Risk or Will Churn / Will Not Churn)

These results help in identifying customers at risk.

- **Data Storage –**

The prediction results are stored in a CSV file (`live_prediction.csv`). This file acts as a data source for the visualization layer and ensures that results are easily accessible.

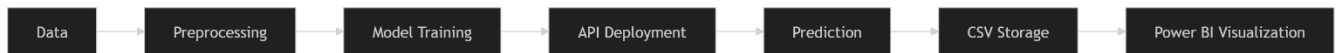
- **Data Visualization (Power BI) –**

Power BI is used to create dashboards that display:

- Churn trends
- Customer segmentation
- Risk levels
- Prediction results

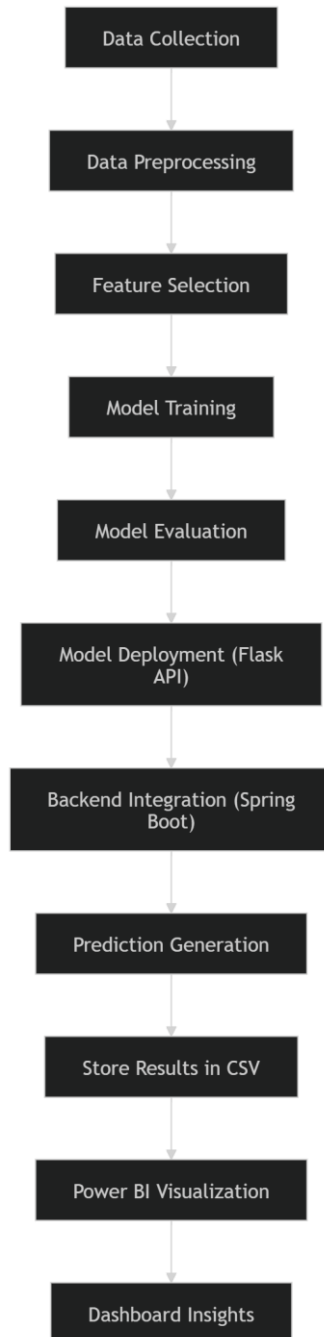
The dashboard reads the updated CSV file and presents insights in a visual format.

- **System Execution Flow –**



- **Alternative Execution (Offline Mode) –**

For demonstration purposes, the system can also run using a Python script (`predict.py`) without backend integration. This ensures that predictions can be generated and visualized even without running the full system.



7. MACHINE LEARNING MODEL –

- **Model Selection Approach –**

In this project, multiple machine learning algorithms were explored to identify the most suitable model for customer churn prediction. Initially, Logistic Regression was

implemented as a baseline model due to its simplicity and effectiveness for classification problems.

- **Use of Logistic Regression (Initial Model) –**

Logistic Regression was first applied to understand the relationship between input features and the churn outcome. It provided a basic prediction model and helped in evaluating how different features influence customer churn. However, since Logistic Regression assumes a linear relationship between variables, it was limited in capturing complex patterns in the dataset.

- **Limitations of Logistic Regression –**

The model showed moderate performance but struggled with:

- Handling non-linear relationships
- Capturing interactions between multiple features
- Providing higher accuracy for complex datasets
- Due to these limitations, an alternative model was considered.

- **Adoption of Random Forest (Final Model) –**

Random Forest was selected as the final model because it is an ensemble learning method that combines multiple decision trees to improve prediction accuracy. It can handle both linear and non-linear relationships effectively.

- **Advantages of Random Forest -**

- Higher accuracy compared to Logistic Regression
- Handles complex data patterns
- Reduces overfitting through ensemble technique
- Works well with both numerical and categorical data

- **Final Model Selection –**

After comparing performance, Random Forest was chosen as the final model for the system. It provided better prediction results and improved reliability.

- **Prediction Output –**

The final model generates:

- A probability score indicating the likelihood of churn
- A classification output (Will Churn / Will Not Churn or High Risk / Low Risk)
- These outputs are used for further visualization and analysis in Power BI.

8. DATASET DESCRIPTION –

- **Overview of the Dataset -**

The dataset used in this project contains customer-related information required to analyze and predict churn behaviour. It consists of multiple records where each record represents a customer and includes various attributes that influence their decision to continue or discontinue the service.

- **Source of Dataset -**

The dataset is obtained from publicly available sources (such as telecom churn datasets from platforms like Kaggle). It is commonly used for classification problems and helps in building predictive models for customer retention.

- **Structure of Dataset -**
The dataset consists of rows (customer records) and columns (features). Each row represents an individual customer, while each column represents a specific attribute of that customer.
- **Target Variable (Dependent Variable) -**
 - **Churn** -
This is the output variable that indicates whether a customer has churned or not.
 - Value = 1 → Customer will churn
 - Value = 0 → Customer will not churn
- **Input Features (Independent Variables) –**
The model uses several important features for prediction:
 - **Tenure (Tenure_Months)** - Represents the number of months a customer has stayed with the company. Customers with shorter tenure are more likely to churn.
 - **Monthly Charges (Monthly_Charges)** - Indicates the amount charged to the customer each month. Higher charges may influence churn behavior.
 - **Total Charges (Total_Charges)** - Represents the total amount paid by the customer over time. It helps in understanding long-term customer value.
 - **Gender (Gender_Male)** - A categorical feature representing the gender of the customer, encoded into numerical form for model processing.
 - **Contract Type** - Indicates the type of service contract:
 - a) Month-to-month
 - b) One year
 - c) Two year.
Customers with long-term contracts are generally less likely to churn.
- **Data Preprocessing Applied –**
Before using the dataset for training, several preprocessing steps were performed:
 - Handling missing values
 - Encoding categorical variables (e.g., gender and contract type)
 - Converting data into numerical format
 - Ensuring consistency across all features
- **Feature Importance –**
Certain features such as tenure and contract type play a significant role in determining churn. These features help the model identify patterns and improve prediction accuracy.
- **Dataset Usage in Project**
 - Training dataset → Used to train the machine learning model
 - Prediction dataset (live_prediction.csv) → Used to store new input and prediction results
 - Visualization dataset → Used by Power BI for dashboard creation
- **Significance of Dataset –**
The dataset forms the foundation of the entire system. Accurate and well-processed data ensures better model performance and reliable predictions.

9. LIBRARIES USED –

- **Pandas** –
Used for data manipulation and analysis. It helps in reading datasets, handling missing values, and preparing data for model training.
- **NumPy** –
Provides support for numerical operations and array processing. It is used for efficient computation during model training and prediction.
- **Scikit-learn (sklearn)** –
A machine learning library used to build and train models. It provides algorithms such as Logistic Regression and Random Forest used in this project.
- **Joblib** –
Used for saving and loading trained machine learning models. It allows reuse of the model without retraining.
- **Flask** –
A lightweight web framework used to deploy the machine learning model as an API. It enables communication between the model and external systems.
- **Flask-CORS** -
Used to handle cross-origin requests, allowing communication between frontend/backend services.
- **Matplotlib / Seaborn** –
Used for data visualization during exploratory data analysis.

10. TOOLS USED –

- **Python (Programming Language) – Visual Studio Code (Flask API Development)**
Used for implementing machine learning models, data processing, and developing the Flask API. Visual Studio Code was used as the primary development environment for writing, running, and debugging the Python code.
- **Power BI** –
Used for creating interactive dashboards and visualizing churn predictions and customer insights.
- **Spring Boot (Java) – IntelliJ IDEA**
Used as the backend integration layer, developed using IntelliJ IDEA to handle API communication and connect different components of the system.
- **Postman** –
Used for testing Flask API endpoints and sending input data to the backend for prediction.
- **IntelliJ IDEA** –
Used as the development environment for building and managing the Java Spring Boot backend application.
- **GitHub** –
Used for version control and storing the project repository.
- **Microsoft Excel** –
Used for viewing and managing dataset files such as CSV or Excel sheets.

11. RESULTS & OUTPUT –

- **Overview of Dataset Results** –
The dataset used in this project contains customer information from a telecom company. It includes various attributes such as customer demographics, service details, and

billing information. The dataset is used to analyze patterns and predict whether a customer is likely to churn.

- **Dataset Characteristics –**

The dataset includes:

- Multiple customer records (rows)
- Various features such as:
 - Gender
 - Senior Citizen
 - Partner and Dependents
 - Tenure (months of service)
 - Phone and Internet Services
 - Contract Type
 - Monthly and Total Charges
 - Churn (target variable)

Each record represents one customer and their service usage details.

- **Target Variable (Churn) –**

The dataset contains a column named **Churn**, which indicates:

- Yes → Customer has left the service
- No → Customer is retained

This column is used as the output variable for training the machine learning model.

- **Model Prediction Results –**

After training the model using the dataset, the system generates:

- **Churn Probability –**
A value between 0 and 1 representing the likelihood of churn
- **Prediction Output**
 - Will Churn (High Risk)
 - Will Not Churn (Low Risk)

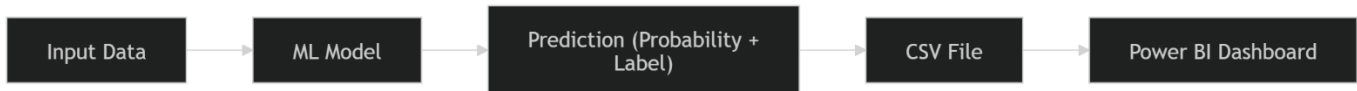
These predictions are stored in `live_prediction.csv` and used for visualization.

- **Power BI Dashboard Output –**

The Power BI dashboard presents the results in an interactive format:

- **Churn Rate Analysis** – Displays the percentage of customers who have churned vs those who have stayed
- **Customer Segmentation** - Shows how churn varies across:
 - Contract types
 - Gender
 - Tenure groups
- **Risk Distribution** - Categorizes customers into high-risk and low-risk groups
- **Prediction Page** - Displays:
 - Input customer details
 - Predicted churn probability
 - Risk classification
 - Insight message (e.g., “Customer likely to churn”)

- **Key Observations from Results –**
 - Customers with **month-to-month contracts** show higher churn rates
 - Customers with **low tenure** are more likely to churn
 - Higher **monthly charges** can increase churn probability
 - Long-term contracts reduce churn significantly
- **System Output Flow –**



- **Effectiveness of Results –**
The system successfully transforms raw customer data into meaningful insights. The combination of machine learning predictions and visualization helps users easily understand customer behaviour and identify potential churners.

12. ADVANTAGES –

- **Improves Customer Retention –**
The system helps businesses identify customers who are likely to churn, allowing them to take preventive actions such as personalized offers and improved services.
- **Data-Driven Decision Making –**
It enables organizations to make informed decisions based on data analysis and predictive insights rather than relying on assumptions.
- **Accurate Prediction –**
The use of machine learning models like Random Forest improves prediction accuracy and helps detect complex patterns in customer behaviour.
- **Interactive Visualization –**
Power BI dashboards provide clear and interactive visualizations, making it easy to understand trends and insights without technical expertise.
- **Scalable System Design –**
The modular architecture allows the system to be extended with additional features such as real-time prediction or web integration.

13. LIMITATIONS –

- **Dependence on Data Quality –**
The accuracy of predictions depends on the quality and completeness of the dataset. Missing or incorrect data can affect results.
- **Not Fully Real-Time –**
The current system updates predictions through CSV files and requires manual refresh in Power BI.

- **Limited Feature Scope** –
The model uses a limited set of features, and additional variables could improve prediction accuracy.
- **Backend Integration Complexity** –
Running both Flask and Spring Boot together may increase system complexity during deployment.

14. FUTURE SCOPE –

- **Real-Time Prediction System** –
The system can be enhanced to support real-time data processing and automatic dashboard updates.
- **Web Application Development** –
A user-friendly web interface can be developed to allow direct input and instant prediction without using external tools.
- **Cloud Deployment** –
The system can be deployed on cloud platforms for better scalability and accessibility.
- **Improved Model Accuracy** –
Advanced algorithms and feature engineering techniques can be applied to improve prediction performance.
- **Integration with CRM Systems** –
The system can be integrated with Customer Relationship Management tools for automated customer retention strategies.

15. CONCLUSION –

The Customer Churn Prediction System successfully demonstrates the application of machine learning and data visualization in solving a real-world business problem. By analyzing customer data and predicting churn probability, the system enables organizations to identify at-risk customers and take proactive measures to improve retention.

The integration of backend technologies and Power BI provides a complete pipeline from data processing to visualization. The use of Random Forest as the final model improves prediction accuracy and ensures reliable results. The interactive dashboard makes it easy for users to understand patterns and insights without requiring technical knowledge.

Overall, the project highlights the importance of combining machine learning with business intelligence tools to support data-driven decision-making. It provides a scalable and flexible solution that can be further enhanced for real-time and enterprise-level applications.

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